

(iv) Every set of independent vectors can be extended to a basis.

Answer: (i),(iii),(iv)

(2) Which of the following statements about the eigendecomposition of a square matrix are FALSE?

(i) Eigenvalues always exist.

(ii) Eigenvalues are always real.

(iii) Eigenvalues are always non-negative.

(iv) Eigenvectors are always orthogonal.

Answer: (ii),(iii),(iv)

$\mathbb{R}$ or $\mathbb{C}$

$\mathcal{V}, \subseteq \mathcal{U}$

(3) Which of the following statements about singular value decomposition are FALSE?

(i) Singular values always exist.

(ii) The left-singular vectors are always orthogonal.

(iii) The right-singular vectors are always orthogonal.

(iv) For a $n \times m$ matrix, singular values are always real.

(v) For a $n \times m$ matrix, singular values are always positive.

Answer: (v)

(4) What is not a property of a linear map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$?

(i) $f(\mathbf{x} + \lambda \mathbf{y}) = f(\mathbf{x}) + \lambda f(\mathbf{y})$

(ii) $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$

(iii) $f(\mathbf{x} \cdot \mathbf{y}) = f(\mathbf{x}) \cdot f(\mathbf{y})$

(iv) $f(\lambda \mathbf{x}) = \lambda f(\mathbf{x})$

Answer: (iii)

(5) Which of the following pairs shows the $(\underline{l_1}, \underline{l_\infty})$ of vector $\begin{pmatrix} 8 \\ 2 \\ 6 \\ -9 \end{pmatrix}$?

(i) (7, 8)

(ii) (7, 9)

(iii) (25, 8)

(iv) (25, 9)

Answer: (iv)

$l_\infty \rightarrow \max_v \text{abs}$

(6) Let $U \subseteq \mathbb{R}^n$ be a subspace. Which of the following are FALSE?

(i) For $\mathbf{u}, \mathbf{v} \in U$, we have $\langle \mathbf{u}, \mathbf{v} \rangle \in U$

(ii) $\mathbf{0} \in U$

(iii) For $\mathbf{u}, \mathbf{v} \in U$, we have $5\mathbf{u} - 8\mathbf{v} \in U$

(iv) For $\mathbf{u} \in U$, we have $3\mathbf{u} \notin U$

Answer: (i), (iv)

(7) Which of the following statements are correct?

(i) All metrics can be induced by norms.

(ii) All norms can be induced by inner products.

(iii) All inner products induce a norm.

(iv) All norms induce a metric.

Answer: (iii), (iv)

(6) Let $U \subseteq \mathbb{R}^n$ be a subspace. Which of the following are FALSE?

(i) For $u, v \in U$, we have $\langle u, v \rangle \in U$ ← inner prod. scalar

(ii) 0 $\in U$

(iii) For $u, v \in U$, we have $5u - 8v$ $\in U$

(iv) For $u \in U$, we have $3u$ $\notin U$

Answer: (i), (iv)

$\lambda u \in U$

(7) Which of the following statements are correct?

(8) Choose all correct statements.

(i) A metric on a set X is a map $d : X \mapsto [0, \infty)$ with the three metric space properties.

(ii) Consider the set $\mathbb{R}$. The map $d(x, y) := \frac{1}{2}|x - y|$ is a metric.

(iii) Consider the set $\mathbb{C}$. The map $d(x, y) := \frac{1}{4}|x - y|$ is a metric.

(iv) For any set X , the map $d : X \times X \mapsto [0, \infty)$ with $d(x, y) = 1$ for $x \neq y$ and $d(x, y) = 0$ for $x = y$ is a metric.

Answer: (ii), (iii), (iv)

(7) Which of the following statements are correct?

(i) All metrics can be induced by norms.

(ii) All norms can be induced by inner products.

~~(iii)~~ All inner products induce a norm.

~~(iv)~~ All norms induce a metric.

Answer: (iii), (iv)

(7) Which of the following statements are correct?

- (i) All metrics can be induced by norms.
- (ii) All norms can be induced by inner products.

(iii) All inner products induce a norm.

(iv) All norms induce a metric.

Answer: (iii), (iv)

$d: X \times X \rightarrow [0, \infty)$

- (i) triangle 'ineq.
- (ii) $d(x, x) = 0$
- (iii) $d(x, y) \geq 0$

(8) Choose all correct statements.

(i) A metric on a set X is a map $d: X \times X \rightarrow [0, \infty)$ with the three metric space properties.

(ii) Consider the set $\mathbb{R}$. The map $d(x, y) := \frac{1}{2}|x - y|$ is a metric.

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Answer: (ii), (iii), (iv)

(9) Consider matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 20 & -10 \\ -1 & -10 & 5 \end{bmatrix}$, please choose all correct statements.

(i) A is symmetric

$(-2)^*$

(ii) A is positive semi-definite (**Hint:** solutions of $ax^2 + bx + c = 0$ are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$)

(iii) tr[A] = 26

(iv) det[A] = 0

Answer: (i), (ii), (iii), (iv)

(10) Let $X = (1, 3] \cup (4, \infty)$ and $d(x, y) = |x - y|$. Which of the following are TRUE?

(i) $(1, 3]$ is closed $\rightarrow X \setminus (1, 3]$ is open

(ii) $(1, 3]$ is open $\rightarrow \varepsilon$ -ball

(iii) Let $C = (1, 3]$, then $\partial C = \{3\}$ $\partial C = \emptyset$

(iv) Let $C = \emptyset$, then C is both open and closed.

Answer: (i), (ii), (iv)

$X \setminus C$

- vector space properties, or a clear example showing why it is not a vector space.
- (b) 3-4 points — Yes/No answer with a partial but mostly correct demonstration of vector space properties.
- (c) 2 points — Yes/No answer that mentions relevant vector space properties but lacks sufficient demonstration or detail.
- (d) 1 point — Yes/No answer only, with no explanation or reference to vector space properties.

(a) $\mathcal{V} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \mid \frac{x_1}{3} = \frac{x_2}{2} = \frac{x_3}{4} \right\}$

Answer: Yes. Assuming

$$\frac{x_1}{3} = \frac{x_2}{2} = \frac{x_3}{4} = z \rightarrow x_1 = 3z, x_2 = 2z, x_3 = 4z$$

hence

$$\mathcal{V} = \left\{ z \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} : z \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \right\}$$

- (i) Zero vector: take $z = 0$ to get $\mathbf{0} \in \mathcal{V}$
- (ii) Closed under addition:

$$\mathbf{u} = z_1 \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}, \mathbf{v} = z_2 \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \rightarrow \mathbf{u} + \mathbf{v} = (z_1 + z_2) \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \in \mathcal{V}$$

- (iii) Closed under scalar multiplication:

$$\alpha \in \mathbb{R} \rightarrow \alpha \mathbf{u} = \alpha z \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \in \mathcal{V}$$

- (i) Zero vector: take $z = 0$ to get $\underline{0} \in \mathcal{V}$
 (ii) Closed under addition:

$$\underline{\mathbf{u}} = \underline{z_1} \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}, \underline{\mathbf{v}} = \underline{z_2} \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \rightarrow \underline{\mathbf{u} + \mathbf{v}} = (\underline{z_1 + z_2}) \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \in \underline{\mathcal{V}}$$

- (iii) Closed under scalar multiplication:

$$\alpha \in \mathbb{R} \rightarrow \underline{\alpha \mathbf{u}} = \underline{\alpha z} \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \in \underline{\mathcal{V}}$$

(b) $\mathbf{V} = \left\{ \mathbf{x} \in \mathbb{R}^n \mid \underline{\mathbf{1}_n^T \mathbf{x}} = \underline{1} \right\}$

0 if $x=0$

Answer: No, for a vector space, zero vector must be in $\mathcal{V}$, but in this example $\mathbf{0} \notin \mathcal{V}$.

(3) [10 points] Eigenvalues of a matrix A .

- (a) If A is a skew-adjoint matrix i.e., $\bar{A} = -A$, then show that all eigenvalues of A lie on the imaginary axis.

Answer: Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be skew-adjoint, i.e. $\mathbf{A}^H = -\mathbf{A}$, and let $\mathbf{Ax} = \lambda \mathbf{x}$ for some eigenpair $(\lambda, \mathbf{x})$ with $\mathbf{x} \neq 0$.

Using the standard Hermitian inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^H \mathbf{u}$, we have

$$\lambda \|\mathbf{x}\|^2 = \langle \mathbf{Ax}, \mathbf{x} \rangle = \mathbf{x}^H \mathbf{Ax}.$$

Also,

$$\lambda^* \|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{Ax} \rangle = (\mathbf{Ax})^H \mathbf{x} = \mathbf{x}^H \mathbf{A}^H \mathbf{x} = \mathbf{x}^H (-\mathbf{A}) \mathbf{x} = -\mathbf{x}^H \mathbf{Ax} = -\lambda \|\mathbf{x}\|^2.$$

Thus $(\lambda + \lambda^*) \|\mathbf{x}\|^2 = 0$. Since $\|\mathbf{x}\|^2 > 0$, we get $\lambda + \lambda^* = 0$, i.e. $\text{Re}(\lambda) = 0$.

Therefore every eigenvalue λ of $\mathbf{A}$ is purely imaginary; equivalently, all eigenvalues of $\mathbf{A}$ lie on the imaginary axis.

- (b) Instead, if $\bar{A}A = I$, then show that all eigenvalues of A lie on the unit circle.

Answer: Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ satisfy $\mathbf{A}^H \mathbf{A} = \mathbf{I}$, and let $\mathbf{Ax} = \lambda \mathbf{x}$ for some eigenpair $(\lambda, \mathbf{x})$ with $\mathbf{x} \neq 0$.

Then

$$\|\mathbf{x}\|^2 = \mathbf{x}^H \mathbf{x} = \mathbf{x}^H \mathbf{A}^H \mathbf{Ax} = (\mathbf{Ax})^H (\mathbf{Ax}) = \|\mathbf{Ax}\|^2 = \|\lambda \mathbf{x}\|^2 = |\lambda|^2 \|\mathbf{x}\|^2.$$

(b) Instead, if $\bar{A}A = I$ then show that all eigenvalues of A lie on the unit circle.

Answer: Let $A \in \mathbb{C}^{n \times n}$ satisfy $A^H A = I$, and let $Ax = \lambda x$ for some eigenpair (λ, x) with $x \neq 0$.

Then

$$\|x\|^2 = x^H x = x^H A^H A x = (Ax)^H (Ax) = \|Ax\|^2 = \|\lambda x\|^2 = |\lambda|^2 \|x\|^2.$$

Since $\|x\|^2 > 0$, it follows that $|\lambda|^2 = 1$, hence $|\lambda| = 1$.

Therefore every eigenvalue of A lies on the unit circle.

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(4) [10 points] Let $T : V \mapsto V$ be a linear transformation that is one-to-one. Show that if $\{v_1, \dots, v_n\}$ is a linearly independent set in V , then so is $\{T(v_1), \dots, T(v_n)\}$.

Hint: Start by considering $c_1 T(v_1) + \dots + c_n T(v_n) = 0$. Show that $c_1 = 0, \dots, c_n = 0$ is the only possible solution.

Answer:

(a) $c_1 T(v_1) + \dots + c_n T(v_n) \xrightarrow{\text{Linearity of } T} T(c_1 v_1 + \dots + c_n v_n) = 0$

(b) Since T is a linear one-to-one transformation: $\ker(T) = \{0\}$

(c) $c_1 v_1 + \dots + c_n v_n = 0$

(d) $c_1 = 0, \dots, c_n = 0$ is the only possible solution since $\{v_1, \dots, v_n\}$ is a linearly independent set in V .

(e) $c_1 T(v_1) + \dots + c_n T(v_n) = 0 \rightarrow \{T(v_1), \dots, T(v_n)\}$ are linearly independent.

Hint: Start by considering $c_1T(v_1) + \dots + c_nT(v_n) = 0$. Show that $c_1 = 0, \dots, c_n = 0$ is the only possible solution.

Answer:

(a) $c_1T(v_1) + \dots + c_nT(v_n) \xrightarrow{\text{Linearity of } T} T(c_1v_1 + \dots + c_nv_n) = 0$

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(d) $c_1 = 0, \dots, c_n = 0$ is the only possible solution since $\{v_1, \dots, v_n\}$ is a linearly independent set in V .

(e) $c_1T(v_1) + \dots + c_nT(v_n) = 0 \rightarrow \{T(v_1), \dots, T(v_n)\}$ are linearly independent.

Step 1: linearity of T

$$c_1T(v_1) + \dots + c_nT(v_n) = 0$$

$$\Rightarrow T(c_1v_1 + c_2v_2 + \dots + c_nv_n) = 0$$

Step 2: Kernel of one-to-one
 $\ker(T) = \{0\}$

$$\text{Step 3: } c_1\underline{v_1} + \dots + c_n\underline{v_n} = \underline{0}$$

$$\text{Step 4: } \underline{c_1} = \underline{c_2} = \dots = \underline{c_n} = 0$$

Step 5: $T(v_1) \dots T(v_n)$ are lin. ind.

(6) (20 points) In this problem, we prove the Cauchy-Schwarz inequality using the concepts learned in the class and then use it to prove the triangle inequality.

(a) (10 points) Prove the Cauchy-Schwarz inequality i.e., $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$.

Hint 1: We may write a vector $\mathbf{x}$ as a scalar multiple of a non-zero vector $\mathbf{y}$, plus a vector orthogonal to $\mathbf{y}$. For example, $\mathbf{x} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} + \left(\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} \right)$

Hint 2: Use the Pythagoras Theorem i.e., $\|\mathbf{x} + \mathbf{y}\|_2^2 = \|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2$ if $\mathbf{x} \perp \mathbf{y}$.

Answer: Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $\mathbf{y} \neq \mathbf{0}$. We can decompose $\mathbf{x}$ into a component parallel to $\mathbf{y}$ and one orthogonal to it:

$$\mathbf{x} = \mathbf{y}_{\parallel} + \mathbf{y}_{\perp}, \quad \text{where} \quad \mathbf{y}_{\parallel} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \mathbf{y}, \quad \text{and} \quad \mathbf{y}_{\perp} \perp \mathbf{y}_{\parallel}.$$

By the Pythagorean theorem,

$$\begin{aligned} \|\mathbf{x}\|^2 &= \|\mathbf{y}_{\parallel}\|^2 + \underbrace{\|\mathbf{y}_{\perp}\|^2}_{\geq 0} \\ &\geq \|\mathbf{y}_{\parallel}\|^2 \end{aligned}$$

Hence,

$$\|\mathbf{x}\|^2 \geq \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2}.$$

Multiplying both sides by $\|\mathbf{y}\|^2$ gives

$$\|\mathbf{x}\|^2 \|\mathbf{y}\|^2 \geq |\langle \mathbf{x}, \mathbf{y} \rangle|^2,$$

and taking the square root of both sides yields the **Cauchy-Schwarz inequality**:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

Hint 1: We may write a vector $\mathbf{x}$ as a scalar multiple of a non-zero vector $\mathbf{y}$, plus a vector orthogonal to $\mathbf{y}$. For example $\mathbf{x} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} + \left(\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} \right)$

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Answer: Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $\mathbf{y} \neq \mathbf{0}$. We can decompose $\mathbf{x}$ into a component parallel to $\mathbf{y}$ and one orthogonal to it:

$$\mathbf{x} = \mathbf{y}_{\parallel} + \mathbf{y}_{\perp}, \quad \text{where } \mathbf{y}_{\parallel} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \mathbf{y}, \quad \text{and } \mathbf{y}_{\perp} \perp \mathbf{y}_{\parallel}.$$

By the Pythagorean theorem,

$$\begin{aligned} \|\mathbf{x}\|^2 &= \|\mathbf{y}_{\parallel}\|^2 + \underbrace{\|\mathbf{y}_{\perp}\|^2}_{\geq 0} \\ &\geq \|\mathbf{y}_{\parallel}\|^2 \end{aligned}$$

Hence,

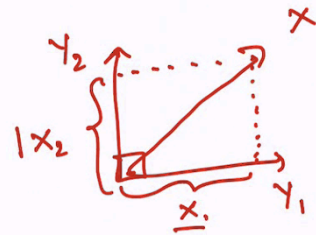
$$\|\mathbf{x}\|^2 \geq \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2}.$$

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and taking the square root of both sides yields the **Cauchy-Schwarz inequality**:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$



$$\mathbf{x} = \alpha \mathbf{y}_1 + \beta \mathbf{y}_2 \quad \left(\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} \right)$$

$$\mathbf{x} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} + \underline{\underline{\mathbf{z}}}$$

$$\|\mathbf{x}\|_2^2 = \frac{(\langle \mathbf{x}, \mathbf{y} \rangle)^2}{\|\mathbf{y}\|_2^2 \|\mathbf{y}\|_2^2} + \|\mathbf{z}\|_2^2$$

and taking the square root of both sides yields the **Cauchy-Schwarz inequality**:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

$$\mathbf{x} = \alpha \mathbf{y}_1 + \beta \mathbf{y}_2 \quad \left(\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} \right)$$

$$\mathbf{x} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|_2^2} \mathbf{y} + \mathbf{z}$$

$$\|\mathbf{x}\|_2^2 = \frac{(\langle \mathbf{x}, \mathbf{y} \rangle)^2}{\|\mathbf{y}\|_2^2} + \|\mathbf{z}\|_2^2 \geq 0$$

$$\|\mathbf{y}\|_2^2 \|\mathbf{x}\|_2^2 \geq (\langle \mathbf{x}, \mathbf{y} \rangle)^2 \Rightarrow \text{Cauchy-Schwarz}$$

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(b) (10 points) Use the Cauchy-Schwarz inequality to prove the triangle inequality i.e., $\|\mathbf{x} + \mathbf{y}\|_2 \leq \|\mathbf{x}\|_2 + \|\mathbf{y}\|_2$.

Answer: We start with the square of the norm:

$$\|\mathbf{x} + \mathbf{y}\|_2^2 = \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle.$$

Expanding the inner product, we get

$$\|\mathbf{x} + \mathbf{y}\|_2^2 = \langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle.$$

By the Cauchy-Schwarz inequality, we have

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2.$$

$\Rightarrow$ apply Cauchy-Schwarz

Hence,

$$\|\mathbf{x} + \mathbf{y}\|_2^2 \leq \|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2 + 2\|\mathbf{x}\|_2 \|\mathbf{y}\|_2 = (\|\mathbf{x}\|_2 + \|\mathbf{y}\|_2)^2.$$

Taking the square root of both sides yields the **triangle inequality**:

$$\|\mathbf{x} + \mathbf{y}\|_2 \leq \|\mathbf{x}\|_2 + \|\mathbf{y}\|_2.$$